Grey Water Washing Machine System – Runbook

Oct 19, 2026 · @Jose

Document control

Item	Detail
Document	Grey Water Washing Machine System – Runbook
Version	3.1
Applies to firmware	`FSM_WashingMachineSystem.ino` and the `GreyWater*` files (firmware 3.1: Spanish and English, chosen in Settings; firmware 3.0: safety task, float-switch rules, code split into modules)
Intended audience	Installers, operators and maintenance personnel
Related document	`README.md` (firmware change summary)

Purpose and scope

This runbook defines how to install, operate, configure, troubleshoot and maintain the grey water controller. It covers the controller firmware and the hardware connected directly to it. The washing machine, the plumbing design and the mains electrical installation are outside its scope.

Safety precautions

- **Isolation:** switch off and lock out the valve and pump supply before working on any wiring. The Stop button is a software control and does not isolate power.
- **Electrical work:** if the valves or pump run on mains voltage, the SSR load wiring must be installed by a qualified electrician to local regulations, with overcurrent protection and an RCD/GFCI.

- **Environment:** mount the controller in an enclosure rated for damp locations, and keep low-voltage wiring separated from mains wiring.
- **Hygiene:** grey water contains detergent residue and bacteria. Wear gloves when servicing the tank, sensors and valves. Stored water is not potable.
- **Supervision:** do not leave the system unattended during commissioning or after a firmware change.

Quick start

For everyday use you only need the two buttons and the screen.

1. **Check the screen** shows `READY` and *Press START*.
2. **Load the washing machine** and start its program as usual.
3. **Press START** (the button labelled START). The screen changes to `1/4 WASH`.
4. **Let it run.** The screen shows the current step, `1/4` to `4/4`. In the last step a countdown shows the time left.
5. **Done.** The screen shows `DONE` and *Cycle complete*, then returns to `READY`.

To cancel at any time, press **STOP** (the button labelled STOP). The screen shows `STOPPED` for 4 seconds.

On a phone connected to the home Wi-Fi, open `http://greywater.local` to see the same status, start or stop the cycle, and check how much water you have saved (section 11).

If the screen shows	It means	What to do
`READY` / Press START	Waiting for you	Press START when the machine is running
`1/4 WASH` … `4/4 SAVE`	Cycle in progress	Nothing; wait
Reusing water	The machine is filling with water saved in the tank	Nothing
Saving water	Rinse water is being stored in the tank for the next wash	Nothing

If the screen shows	It means	What to do
`DONE` / Reused 26 L	Cycle finished; that much tap water was saved	Nothing
`STOPPED` / Cycle stopped	You pressed STOP	Press START to begin again
No water flow	A step has waited too long for water	Press STOP; check the Service tab of the app, see section 8
Blank screen	No power, or a display fault	Check power, then see section 8

Button colors are a recommendation; label your buttons to match.

1. Overview

The controller runs a 4-step wash cycle (Fill, Drain, Rinse, Drain), reuses grey water from a storage tank for the first fill, and saves the rinse water back to the tank. It runs on an Arduino Nano ESP32, drives five solid-state relays (SSRs), reads two flow sensors and two tank level switches, and reports status on a 128x32 OLED.

Components

Group	Device	What it does	Board pin	Part / type
Controller	Arduino Nano ESP32	Runs the program	—	ABX00083 (ESP32-S3, 3.3 V logic)
Operator	START button	Starts a cycle	D11	Momentary push button, normally open
Operator	STOP button	Cancels a cycle	D10	Momentary push button, normally open
Operator	OLED screen	Shows status and progress	A4 (SDA), A5 (SCL)	SSD1306, 128x32, I2C 0x3C
Sensors	Tap flow sensor	Detects water entering the machine	D2	YF-B1

Group	Device	What it does	Board pin	Part / type
Sensors	Drain flow sensor	Detects water leaving the machine	D3	YF-B10
Sensors	Tank low-level switch	Water at the low mark	D9	Float switch
Sensors	Tank high-level switch	Water at the high mark (tank full)	D8	Float switch
Actuators	Tap valve	Tap water supply	D6	Solenoid valve via SSR
Actuators	Drain valve	Drain path from the machine	D4	Solenoid valve via SSR
Actuators	Fill valve	Sends drain water into the tank	D5	Solenoid valve via SSR
Actuators	Pump	Moves tank water to the machine	D7	Pump via SSR
Actuators	Tank-wash valve	Washes the tank during the tank-water fill	D0 / RX0	Solenoid valve via SSR

Wash cycle summary

1. **Fill (1/4):** the machine fills with tap water, or with tank water when the tank has water and the machine is demanding water. The tank-wash valve pulses every 10 s while tank water is used.
2. **Drain (2/4):** soapy water is drained, not saved.
3. **Rinse (3/4):** the machine fills with tap water.
4. **Drain (4/4):** rinse water is saved to the tank until it is full. This step lasts 20 minutes, then the system returns to Idle.

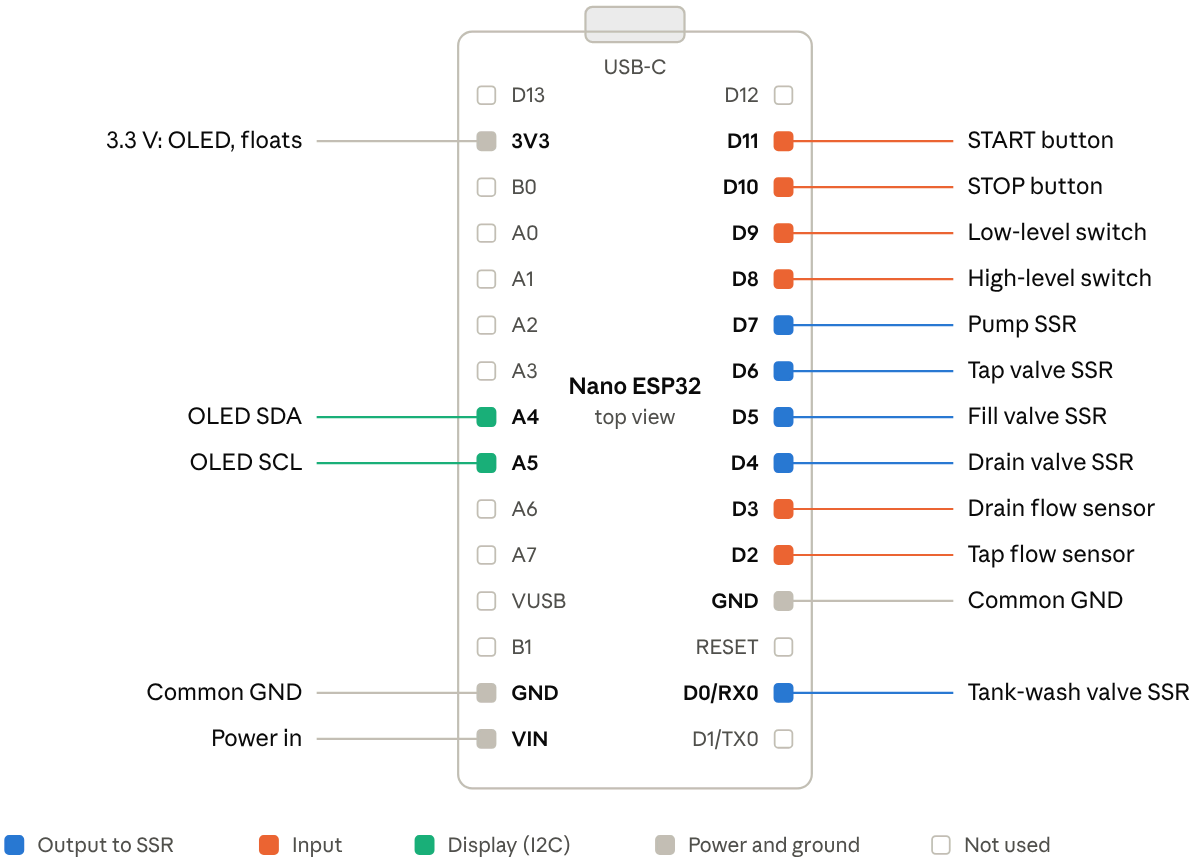
Source files (from firmware 3.0): `FSM_WashingMachineSystem.ino` (setup and loop), `GreyWaterSystem` (wash cycle), `GreyWaterOutputs` and `GreyWaterTask` (outputs, float switches, safety task), `GreyWaterSensors`, `GreyWaterDisplay`, `GreyWaterDiagnostics`, `GreyWaterSettings`, `GreyWaterNet`, `FlowMeter.h`, `TankLevel.h`, and the app page `web/dashboard.html` with its Spanish texts `web/es.json`. Libraries: Adafruit GFX, Adafruit SSD1306, AgileStateMachine, WiFiManager.

2. Pinout configuration

All pins are defined in `GreyWaterSystem.h` and set up in `setup()`. Outputs are active HIGH (`ssrON = HIGH`); buttons are active LOW with internal pull-ups.

Board pinout

Sensors, buttons and valves use the D0–D11 header; the display uses A4 and A5



Arduino Nano ESP32 pinout · pins used by this system

Filled squares are the pins in use. Match each pin by the label printed on the board, not by its position, because the drawing is not to scale.

Board pin	ESP32-S3 GPIO	Connect to	Type	Reads / drives	Firmware name
Inputs					
D2	GPIO5	Tap flow sensor signal	Pulse input (interrupt)	Pulses while water flows	`TAP_PIN`

Board pin	ESP32-S3 GPIO	Connect to	Type	Reads / drives	Firmware name
D3	GPIO6	Drain flow sensor signal	Pulse input (interrupt)	Pulses while water flows	`DRAIN_PIN`
D8	GPIO17	Tank high-level switch	Digital input	HIGH = water at the high mark	`IN_HIGHLEVEL`
D9	GPIO18	Tank low-level switch	Digital input	HIGH = water at the low mark	`IN_LOWLEVEL`
D10	GPIO21	STOP button (other side to GND)	Input, internal pull-up	LOW = pressed	`STOP_BUTTON`
D11	GPIO38	START button (other side to GND)	Input, internal pull-up	LOW = pressed	`START_BUTTON`
Outputs					
D0 / RX0	GPIO44	Tank-wash valve SSR, input +	Digital output (shared with UART0 RX)	HIGH = valve on	`OUT_WASH_EV`
D4	GPIO7	Drain valve SSR, input +	Digital output	HIGH = valve on	`OUT_DRAIN_EV`
D5	GPIO8	Fill valve SSR, input +	Digital output	HIGH = valve on	`OUT_FILL_EV`
D6	GPIO9	Tap valve SSR, input +	Digital output	HIGH = valve on	`OUT_TAP_EV`
D7	GPIO10	Pump SSR, input +	Digital output	HIGH = pump on	`OUT_MOTOR`
Display					
A4	GPIO11	OLED SDA	I2C data, 400 kHz	—	`Wire`
A5	GPIO12	OLED SCL	I2C clock, 400 kHz	—	`Wire`
Power					
3V3	—	OLED VCC and level-switch common	3.3 V supply out	—	—
GND	—	Common ground: buttons, sensors, OLED, every SSR input −, and the 5 V supply −	Ground	—	—
VIN	—	Controller supply (see the board datasheet for the allowed voltage)	Power in	—	—

Wiring requirements

- **SSR inputs:** connect each output pin to the SSR's DC input (+) and GND to (−). The board must not drive solenoids or the motor directly.
- **Level switches:** D8 and D9 use plain `INPUT`, so each switch needs an external pull-down (or pull-up, with the logic inverted in `getWaterLevelTank()`). A floating input gives random tank levels.
- **Buttons:** wire between the pin and GND; no external resistor is needed.
- **Flow sensors:** power them at the voltage the sensor and the board's inputs both accept. The Nano ESP32 inputs are 3.3 V only and not 5 V tolerant. If a sensor is powered from 5 V, use the diode and 10 kΩ pull-up shown in the wiring diagrams on D2 and D3. Do not use a resistor divider: with the sensor's own pull-up it can leave the signal just above the pin's switching level, so pulses drop out.
- **Pin D0 (tank-wash valve):** D0 is also the board's RX0 serial pin. The sketch's `Serial` uses the USB port, so D0 is free, but do not use `Serial0` in the same sketch. Fit a 10 kΩ resistor from D0 to GND so the SSR stays firmly off while the board starts. Always write `D0` in the code, not a bare `0`.
- **Boot state:** `allSSROff()` writes LOW to all five outputs in `setup()`, so every valve and the motor start off.

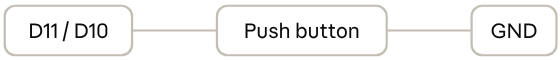
Pin numbers follow the Arduino labels printed on the board (Arduino IDE: **Tools > Pin Numbering > By Arduino pin (default)**). The GPIO column is for reference only. Source: Arduino Nano ESP32 full pinout.

Wiring diagrams

Each panel shows one type of device; wire every device of that type the same way, using the pin from the pinout table.

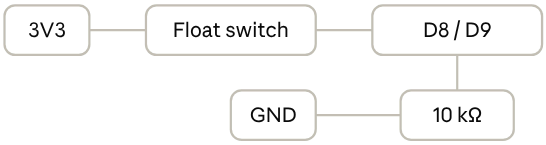
Sensors and buttons run at 3.3 V; only the SSR load side carries valve power

START and STOP buttons (D11, D10)



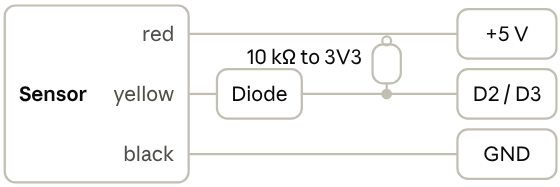
Uses the internal pull-up: no resistor needed.
Pressed = LOW. Use normally open buttons.

Tank level switches (D8, D9)



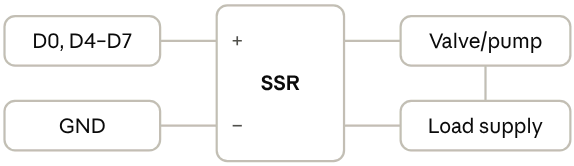
Float up: switch closed, pin reads HIGH.
The 10 kΩ pull-down holds the pin LOW when open.

Flow sensors (D2, D3)



Diode + 10 kΩ pull-up: a clean 0–3.3 V signal.
Diode cathode to the sensor; colors are typical.

Valves and pump via SSR (D0, D4–D7)



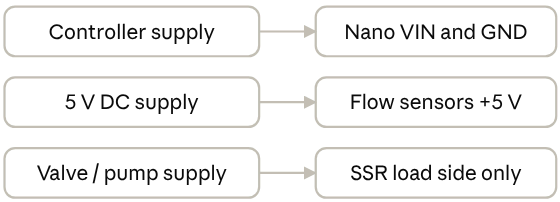
AC valves: AC-output SSR on the load supply.
DC valves: DC-output SSR and a flyback diode.

OLED display (A4, A5)



Address 0x3C; most modules include pull-ups.

Power supplies and ground



Join low-voltage GNDs; keep load supply separate.

wiring diagrams · one circuit per device type

Device	Wire 1	Wire 2	Wire 3	Extra parts
START / STOP button	Pin D11 / D10	GND	—	None
Level switch (×2)	3V3	Pin D8 / D9	—	10 kΩ from the pin to GND

Device	Wire 1	Wire 2	Wire 3	Extra parts
Flow sensor (×2)	Red: +5 V supply	Black: GND	Yellow: pin D2 / D3 through a Schottky diode (cathode to the sensor)	10 kΩ from the pin to 3V3; optional 1–10 nF from the pin to GND
SSR (×5)	Input +: pin D0 or D4–D7	Input −: GND	—	Load side in series with the valve or pump and its supply
OLED	VCC: 3V3	GND: GND	SDA: A4, SCL: A5	None

Choose SSRs whose control input switches reliably at 3.3 V (for example, a 3–32 V DC input). Use an AC-output SSR for AC valves and pumps, and a DC-output SSR with a flyback diode across the coil for DC valves.

3. Outputs and relay states

From firmware 3.0, only `GreyWaterOutputs` switches the outputs. The active step requests one of the modes below every 250 ms (`OPERATION_DELAY`), and a separate safety task checks the floats and the pump every 20 ms, whatever the rest of the program is doing (for example while a phone loads the app).

What the system is doing	When	Tap valve (D6)	Drain valve (D4)	Fill valve (D5)	Pump (D7)	Tank-wash valve (D0)	Firmware function
Starting up	Power-on	Off	Off	Off	Off	Off	`allSSROff()`
Waiting / not saving	Idle, Run, step 4 with tank full	**On**	**On**	Off	Off	Off	`noSavingWater()`
Washing the tank manually	Idle, after Wash tank on the dashboard	**On**	**On**	Off	Off	**On** until the tank is full	`idleStateLogic()`

What the system is doing	When	Tap valve (D6)	Drain valve (D4)	Fill valve (D5)	Pump (D7)	Tank-wash valve (D0)	Firmware function
Filling with tap water	Step 1 (tap), step 3	**On**	**On**	Off	Off	Off	`fillWashingMachineWithTapWater()`
Filling with tank water	Step 1 (tank), while the machine draws water	Off	**On**	Off	**On**; off within 0.3 s of the machine closing its inlet	**Pulsed**: 2 s every 10 s	`fillWashingMachineWithTankWater()`
Draining soapy water	Step 2	**On**	**On**	Off	Off	Off	`drainSoapyWater()`
Saving water to the tank	Step 4, tank not full	**On**	Off	**On**	Off	Off	`checkAndFillWaterTank()`

Pump protection (from firmware 2.5): the pump runs only while water is actually flowing.

- **Fast cut-off:** if no tap-sensor pulse arrives for 0.3 s, the pump stops and the tap valve opens at once. That happens when the machine closes its inlet, or when the pump delivers nothing (air lock, blocked intake, empty pipe). The pump gets 1 s after starting for water to flow.
- **No restart into a closed inlet:** after a cut-off, the pump waits for the machine's next draw.
- **Alert:** after 3 starts with no water in one cycle, the controller sends a *tank pump not pumping* alert.
- **Hardware limit:** firmware cannot remove the last 0.3 s. A pump with a built-in pressure switch, or a bypass back to the tank, protects it fully.

Interlocks (from firmware 3.0): applied every 20 ms in every state.

- The fill valve (D5) and the tank-wash valve (D0) never open while the high float is wet. In step 4 the rinse water then goes to the drain, and saving stays off for the rest of that step.
- The pump (D7) never runs while the low float is dry; the tap supplies the machine instead.

- The pump cut-off above keeps working even if the main program is busy.

Tank level decoding

From firmware 3.0, the safety task reads both switches every 20 ms. Each switch reports its safe reading after 0.2 s (high float wet, low float dry) and the other reading only after 2 s, so a float bobbing at the mark does not switch valves on and off:

Low switch D9	High switch D8	Level	OLED tank icon
LOW	LOW	low	Empty
HIGH	LOW	medium	Half full
HIGH	HIGH	high	Full
LOW	HIGH	high: the tank is treated as full and the pump is not used. After 3 s it is reported as a switch fault (alert *check level switches*, app *Check float switches*); see section 8	Full

Tank-wash timing

While `fillWashingMachineWithTankWater()` runs, the wash valve opens immediately, stays open for `TANK_WASH_OPEN_TIME` (2 s), and reopens every `TANK_WASH_PERIOD` (10 s). The timer restarts on every state entry. Switching to tap water closes the valve; switching back to tank water in the same state continues the same 10 s cycle.

4. Buttons and operator controls

The system has two momentary buttons: Start begins a cycle from Idle, and Stop ends a cycle from any other state. Both are read every `loop()` pass by `updateInputs()` with a 20 ms debounce (`DEBOUNCE_DELAY`).

Button	Pin	Press it when	What happens	Screen shows	Ignored when
START	D11	The screen shows `READY` and the washing machine program is running	The cycle begins; step 1 starts after 250 ms	`STARTING`, then `1/4 WASH`	A cycle is already running

Button	Pin	Press it when	What happens	Screen shows	Ignored when
STOP	D10	You want to cancel the cycle	The system returns to Idle and resets all timers	`STOPPED` for 4 s, then `READY`	The system is already in Idle

Operational notes

- **Start in any other state:** ignored. A running cycle cannot be restarted without pressing Stop first.
- **Stop in Idle:** ignored.
- **Stop paths:** from Run and Fill 1/4 the state machine's own transition goes to Idle. From Drain 2/4, Rinse 3/4 and Drain 4/4, `loop()` also forces a reset with `resetFSMToIdle()`. Both end in the same Idle entry.
- **Buttons act while held:** the debounce reports the button as pressed for as long as it is held, not once per press. Holding Start when a cycle ends starts the next cycle straight away.
- **Both buttons at once:** do not press both buttons simultaneously. The resulting state depends on the order in which transitions are evaluated.
- **Stop is not an emergency stop.** In Idle the tap and drain SSRs stay on (`noSavingWater()`). To make the installation safe, isolate the SSR and valve supply (see Safety precautions).

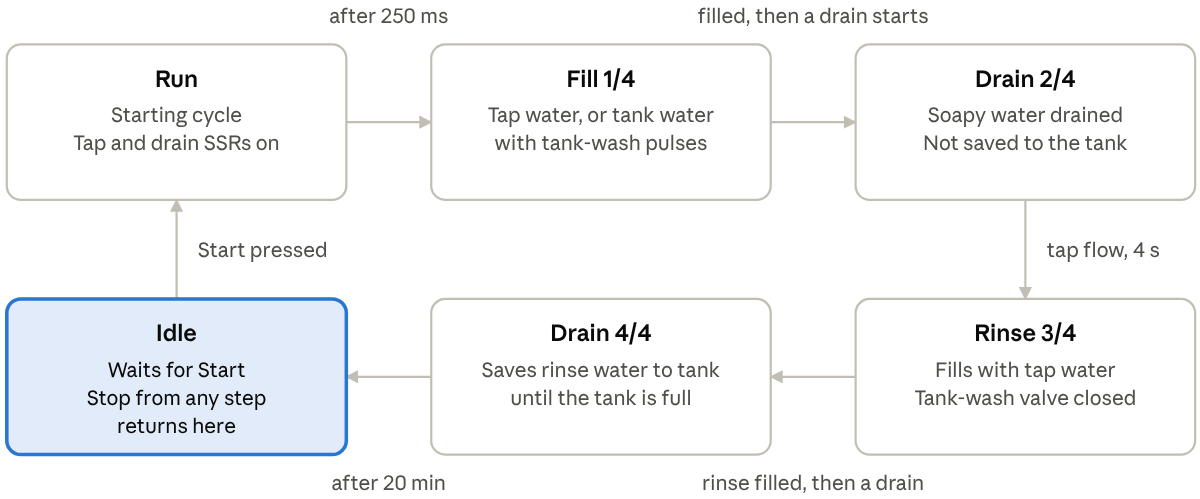
Standard operating procedure

1. Check that the OLED shows `READY` / Press START.
2. Load the washing machine and start its program.
3. Press Start. The header changes to `1/4 WASH`.
4. No operator action is required while the cycle runs. Progress shows in the header (`1/4` to `4/4`); in step 4 a countdown shows the time left.
5. When the OLED shows `DONE` / Cycle complete, the system is back in Idle.
6. To abort, press Stop at any time.

5. Operating procedure

Pressing Start runs the four steps in order; each of steps 1–3 ends only when the flow sensors confirm what the washing machine is doing, and step 4 ends after 20 minutes.

Start runs four steps, then the system returns to Idle



state machine · 6 states, Stop returns to Idle from every step

The diagram reads clockwise from Run. "Flow, 4 s" means the flow rate stayed above the Flow threshold (default 1 L/min) for the flow detection time, `EDGE_DETECTION_DELAY` (4 s).

State	On entry	Every 250 ms	Next state	Exit condition
Idle	Clears buttons, cycle flags, edge flags, flow counters, tank-wash timer; shows the result notice	`noSavingWater()`	Run	Start pressed
Run	Clears any notice	`noSavingWater()`	Fill 1/4	250 ms elapsed

State	On entry	Every 250 ms	Next state	Exit condition
Fill 1/4	Resets flow counters and tank-wash timer	`checkAndFillWashingMachine()`: tap water if the tank is low or no demand yet, else tank water with wash pulses	Drain 2/4	The machine is filled (took at least the Minimum fill, default 3 L, and its inlet closed), then a drain starts (`filled1`). A drain before or during the fill does not count.
Drain 2/4	Resets flow counters	`drainSoapyWater()`	Rinse 3/4	Tap flow seen for 4 s: the machine starts its rinse fill (`drained1`)
Rinse 3/4	Resets flow counters	`fillWashingMachineWithTapWater()`	Drain 4/4	The rinse fill is complete (Minimum fill taken, inlet closed), then a new drain starts: the rinse drain (`filled2`)
Drain 4/4	Resets flow counters; starts the countdown	`checkAndFillWaterTank()`: save to tank while low or medium, else not saving	Idle	`TRANSITION_DELAY` (20 min) elapsed

Steps 1–3 have no timeout. If the expected flow never arrives, the step waits until Stop is pressed (section 8).

The controller follows the washing machine; it does not command it. Start the machine's own program, then press Start, and keep the machine's program length in line with `TRANSITION_DELAY`.

6. OLED display guide

The screen shows the step in the header, two live indicators on the right, and a one-line message below.

```
+--------------------------------+
|1/4 WASH               [##] [W] |   header: step + tank level + wash valve
|--------------------------------|
|          Reusing water         |   message: short, fits without scrolling
+--------------------------------+
```

Element	Reads	Meaning
Header text	`1/4 WASH` etc.	Current step; in step 4 also the time left (`4/4 SAVE 18:42`)
Tank icon	Empty / half / full box	Tank level from the level switches (low / medium / high)
`W` icon	Outlined / white box	Tank-wash valve closed / open
Message	FreeSans 9 pt	What the system is doing now

Messages

Top line	Message	What it means	Action needed
`GREY WATER`	System ready	The controller has just switched on (2 s)	None
`READY`	Press START	Waiting for a new cycle	Press START when the machine runs
`STARTING`	Starting…	START was pressed	None
`1/4 WASH`	Waiting to fill	The machine has not started taking water yet	Start the machine’s program if it is not running
`1/4 WASH`	Reusing water	The machine is filling with water saved in the tank; the tank is being cleaned (`W` lights every 10 s)	None

Top line	Message	What it means	Action needed
`1/4 WASH`	Using tap water	The machine is filling from the tap	None
`1/4 WASH`	Tank low: tap	The tank has too little water, so tap water is used	None
`1/4 WASH`	Washing...	The machine has filled and is washing	None
`2/4 SOAPY DRAIN`	Not saving	Soapy wash water goes to the drain, not the tank	None
`3/4 RINSE`	Using tap water	Rinse fill from the tap	None
`3/4 RINSE`	Rinsing...	The machine has filled and is rinsing	None
`1/4`, `2/4` or `3/4`	No water flow	The step has waited longer than the step time limit for water flow	Check the Service tab of the app
`4/4 SAVE 18:42`	Saving water	Rinse water is stored; the clock shows the time left	None
`4/4 SAVE 18:42`	Tank full: drain	The tank is full; the rest goes to the drain	Use or empty the tank if this happens often
`TANK WASH 9:42`	Cleaning tank	Manual tank wash; the clock shows the time left before the safety limit	None
`TANK CLEANED`	Tank full	The manual wash filled the tank (4 s)	None
`WASH STOPPED`	Stopped early	Stopped from the app, the STOP button, or by a cycle starting (4 s)	None
`WASH STOPPED`	Check tank	The time limit was reached, or the level switches disagree	Check both float switches and the wash water supply

Top line	Message	What it means	Action needed
`GreyWater-Setup`	Join this Wi-Fi	The Wi-Fi setup page is open; the top line is the network to join	Join it on a phone, password `greywater`
IP address	Connected	Connected to the home Wi-Fi; the top line is the app's address (6 s)	Note it, or use `greywater.local`
`UPDATING`	Keep power on	A wireless firmware update is running	Wait; do not switch off
`STOPPED`	Cycle stopped	STOP was pressed (4 s)	Press START to run again
`DONE`	Reused 26 L	The cycle finished; tank water replaced that much tap water (4 s). Shows Cycle complete if none was reused	None

Every message fits the screen without scrolling; each was measured in the screen's font. The app uses the same words in full sentences.

The controller's screen uses short messages that fit without scrolling. The dashboard shows the same status as full sentences.

Short messages are centred and still. Messages wider than 128 px pause for 1.5 s, then scroll at 50 px/s and repeat. The display only sends a new frame when something changes or while text scrolls.

7. Configuration parameters

All timing and behavior settings are `#define`s in `GreyWaterSystem.h`; change them there and re-flash.

Constant	Default	Unit	Controls	Guidance
`OPERATION_DELAY`	250	ms	How often each state updates relays and sensors	Keep 100–1000; other timings are rounded to this step
`TRANSITION_DELAY`	1200000	ms	Length of step 4 (20 min) before returning to Idle	Set to how long the machine's rinse drain lasts
`EDGE_DETECTION_DELAY`	4000	ms	Flow must last this long to count as "flow detected"	Raise if short spurts end steps early
`TANK_WASH_PERIOD`	10000	ms	Interval between tank-wash openings	Multiple of `OPERATION_DELAY`
`TANK_WASH_OPEN_TIME`	2000	ms	Wash valve open time per interval	Less than `TANK_WASH_PERIOD`, else always open
`DEBOUNCE_DELAY`	20	ms	Button debounce	20–50
`CONTRL_SSR_DELAY`	0	ms	Blocking delay after each SSR write	Keep 0; any value blocks `loop()` 5× per update
`DISPLAY_FRAME_MS`	40	ms	Frame interval while scrolling (25 fps)	Not below about 20
`SCROLL_SPEED_PX`	2	px/frame	Scroll speed (50 px/s)	1–3
`SCROLL_PAUSE_MS`	1500	ms	Rest before a long message scrolls	—

Constant	Default	Unit	Controls	Guidance
`SCROLL_GAP_PX`	40	px	Gap before a scrolled message repeats	—
`NOTICE_DURATION`	4000	ms	How long "Cycle stopped" / "Cycle complete" stays	—
`DEBUG_MODE`	false	—	Serial debug output at 9600 baud	Set `true` for troubleshooting
`SENSOR_TAP_TYPE`	YFB1	—	Tap flow sensor model	Must match the fitted sensor
`SENSOR_DRAIN_TYPE`	YFB10	—	Drain flow sensor model	Must match the fitted sensor

The SSR logic level is set in the `.ino` (`ssrON = HIGH`, `ssrOFF = LOW`). For active-LOW relay boards, swap the two values.

From firmware 2.0, the wash timing, final drain length, flow detection time and alert limits are changed on the dashboard (section 11) and saved in the controller. The `#define` values are only the factory defaults used on first boot.

8. Exceptions and troubleshooting

The firmware handles a low or full tank automatically, but it has no timeouts on steps 1–3 and no fault alarms, so a missing flow signal leaves the cycle waiting until Stop is pressed.

Conditions handled by the firmware

Condition	Detected by	Automatic response	OLED
Tank low during Fill 1/4	Level switches	Fills from the tap instead of the tank	Tank low: using tap water

Condition	Detected by	Automatic response	OLED
No tap demand yet in Fill 1/4	Tap flow sensor	Stays on tap water until flow is confirmed	Using tap water
Tank full during Drain 4/4	High-level switch	Stops saving; water goes to the normal drain path	Tank full: not saving
High float wet, low float dry for 3 s (from 3.0)	Both level switches	Tank treated as full: no saving, no tank wash, no pump; tap water used. High-priority alert, alarm beeps, event logged	App: Check float switches
Stop pressed	Stop button	Returns to Idle, clears all flags and timers	Cycle stopped
Power loss or reset	—	Restarts in Idle with all outputs off; the cycle is not resumed	System ready, then Press START
`millis()` rollover after 49.7 days	—	Timers use unsigned subtraction, so they keep working	—

Faults not detected by the firmware

Problem	Most likely cause	What to check	How to fix	Who
Screen blank, nothing responds	Display not detected; the program stops with all outputs off	Power to the board; OLED wires on 3V3, GND, A4, A5	Reseat the OLED wiring; confirm address 0x3C	Technician
Stuck on `1/4 WASH`	One flow sensor reports no water	Washing machine program is running; both flow sensors (D2, D3)	Press STOP; clean or reconnect the sensor	User, then technician
Stuck on `2/4 SOAPY DRAIN`	Tap flow sensor reports no water	Tap valve opens; tap sensor on D2	Press STOP; check the tap sensor	Technician

Problem	Most likely cause	What to check	How to fix	Who
Stuck on `3/4 RINSE`	Drain flow sensor reports no water	Drain hose and drain sensor on D3	Press STOP; check the drain sensor in the dashboard's Controller panel (pulses counting with no water = noise: fix the wiring); raise the Flow threshold if a steady trickle continues	Technician
Steps change too early	Electrical noise or short spurts of water	Sensor cable routing near mains wires	Separate the sensor cables from mains, pump and valve wiring; fit the diode and pull-up interface; raise the Flow threshold in Settings	Technician
Tank icon wrong	Level switch stuck, or a missing pull-down resistor	Move each float by hand and watch the icon	Clean the floats; fit the 10 kΩ pull-downs	Technician
Fill keeps switching tap ↔ tank	Float bouncing or water sloshing	Float mounting	Mount the float in a calm spot or a still tube	Technician
`W` lights but no tank wash	SSR wired to the wrong pin, or no valve supply	SSR on D0; SSR LED; valve supply	Rewire to D0; restore supply	Technician
Tank-wash valve open the whole fill	`TANK_WASH_OPEN_TIME` is not less than `TANK_WASH_PERIOD`	Settings in `GreyWaterSystem.h`	Set the open time below the period	Technician

Problem	Most likely cause	What to check	How to fix	Who
A valve or the pump never switches	Failed SSR, or SSR needs a different logic level	Voltage on the pin (3.3 V when on); SSR LED	Replace the SSR; swap `ssrON` / `ssrOFF` for active-LOW boards	Technician
Screen stutters, buttons slow	I2C too slow, or a delay added to SSR switching	`Wire.setClock(400000)`; `CONTRL_SSR_DELAY`	Keep 400 kHz and a 0 ms delay	Technician
A new cycle starts by itself	START button held or stuck	Button and its wiring	Release or replace the button	User

Diagnostic procedure

1. Set `DEBUG_MODE` to `true` in `GreyWaterSystem.h` and flash.
2. Open the Serial Monitor at 9600 baud.
3. Watch the state entries (`Entering state: …`), flow rates, tank level and SSR switching while you reproduce the fault.
4. Set `DEBUG_MODE` back to `false` for normal use; the debug output slows the loop.

Developer note: `DEBUG_PRINT(x)` expands to an `if` without braces. Do not use it as the only statement before an `else`.

9. Commissioning and test checklist

Run these checks after installation, after any wiring change, and after flashing new firmware.

Before power-up

- [ ] Every SSR input is wired to its pin from section 2, with a common GND
- [ ] SSR load side and valve/motor supply are fused and switched off
- [ ] Level switches have pull resistors; buttons go to GND
- [ ] Flow sensor signal levels are safe for the board's inputs

Power-up and idle

- [ ] OLED shows `GREY WATER` / System ready for 2 s, then `READY` / Press START, with no "Cycle complete"
- [ ] All SSRs are off for the first moment, then tap and drain SSRs turn on in Idle
- [ ] Tank icon follows the level switches when each float is lifted by hand (empty, half, full)

Buttons

- [ ] Start in Idle changes the header to `STARTING`, then `1/4 WASH`
- [ ] Stop during each step shows `STOPPED` / Cycle stopped for 4 s, then `READY`
- [ ] Start during a running step has no effect

Full cycle

- [ ] Fill 1/4 with the tank low uses tap water; the `W` icon stays outlined
- [ ] Fill 1/4 with water in the tank switches to tank water once the machine draws water; the motor runs and `W` turns white for 2 s every 10 s
- [ ] Drain 2/4 shows Draining soapy water; the fill valve stays off
- [ ] Rinse 3/4 uses tap water; the wash valve stays closed
- [ ] Drain 4/4 saves water (fill valve on, drain off) and the countdown decreases each second
- [ ] With the tank full, Drain 4/4 shows Tank full: not saving
- [ ] After 20 minutes the OLED shows `DONE` / Cycle complete, then `READY`

Faults

- [ ] Disconnect the OLED and reset: all outputs stay off
- [ ] Disconnect a flow sensor in Fill 1/4: the step waits, and Stop still returns to Idle

10. Maintenance and firmware deployment

Maintenance

Task	Interval	Notes
Check the OLED tank icon against the real tank level	Monthly	Catches stuck or fouled level switches
Clean flow sensors and their inlet screens	Every 3 months	Lint in grey water blocks the YF-B10 drain sensor first
Inspect the tank and its wash nozzle	Every 3 months	Adjust `TANK_WASH_OPEN_TIME` if sediment builds up
Check valves and SSRs switch (run the commissioning "Full cycle" checks)	Every 6 months	Listen for each valve click; check SSR LEDs
Check connectors and cable strain relief	Every 6 months	Damp environment

Intervals are suggested starting points; adjust them to the water quality and how often the machine is used.

Firmware build and installation

1. Install the Arduino IDE (or arduino-cli) and the board package for your ESP32-based board.
2. Install the libraries: Adafruit GFX, Adafruit SSD1306, AgileStateMachine and WiFiManager (from firmware 2.4 the FlowSensor library is no longer needed).
3. Put `FSM_WashingMachineSystem.ino`, `GreyWaterSystem.cpp` and `GreyWaterSystem.h` in one folder named `FSM_WashingMachineSystem`.
4. Select the board and port, then Upload.
5. Run the checks in section 9.

Software modification reference

- Pins, timings and display settings: `GreyWaterSystem.h` (sections 2 and 7).

- Relay combinations: the preset functions near `setRelayState()` in `GreyWaterSystem.cpp` (section 3).
- OLED text: `displaySetStatus()` calls in each state's logic function, and the strings returned by `checkAndFillWashingMachine()` and `checkAndFillWaterTank()`.
- States and transitions: `setupStateMachine()` in the `.ino` (section 5).

11. Wi-Fi features (firmware 2.0)

Firmware 2.0 adds a phone dashboard, alerts, water statistics and wireless updates; the wash cycle runs exactly the same when Wi-Fi is missing.

First-time Wi-Fi setup

1. On first boot the screen shows `GreyWater-Setup` and the setup key, for example `Key gw-a1b2c3` (in Spanish, only the key). The key is unique to each board (from firmware 3.0) and is also printed on the USB serial monitor.
2. On a phone, join the Wi-Fi network **GreyWater-Setup** with that key. Only if the screen is missing or broken is the key the old shared password `greywater`.
3. A setup page opens (otherwise browse to `192.168.4.1`). Choose the home network and enter its password.
4. The screen shows the controller's IP address for 6 seconds.
5. Open `http://greywater.local` (or that IP address) and sign in as `admin`, password `greywater`.
6. In **Settings**, set new passwords and an alert topic (see Phone alerts).

The setup page closes after 3 minutes and the system then runs offline. To open it again, for example after changing the router, hold **STOP for 5 seconds** while the screen shows `READY`.

From firmware 3.0 the setup page only sets the Wi-Fi network and shows information: it cannot install firmware or erase the controller, because it has no password of its own.

Dashboard

Tab	For	Shows
Overview	Household	What is happening now, using the same words as the controller's screen; Start cycle and Stop cycle; the storage tank (level, water age, wash valve) and the Wash tank button
Water use	Household	Last wash (when, how long, litres reused and saved); this month's share of washing water that came from the tank, compared with last month; washes, reused, saved and tap water this month; totals by period
Service	Installers and technicians	Flow sensors (rate, flowing or not, pulse counts); valves and pump; tank level switches; current step with litres taken and the "machine is filled" guard; Wi-Fi, IP address, uptime, alerts and firmware
Settings	Everyone	All the values in the settings table, a test-alert button and a restart button

On the Service tab, the tank-wash valve state is reported by the controller. The other valve and pump states are the expected states for the current step and fill source.

Refresh (from firmware 2.8.1): the app updates itself every 2 seconds. The badge in the top bar shows the connection:

- *Online* with the controller's time: updates are arriving.
- *Updating slowly*: no update for 6 seconds, usually weak Wi-Fi.
- *Offline*: the controller cannot be reached.

Press **Refresh** next to the badge to fetch fresh data at once. It cancels any update that is still waiting.

Keeping the water history

The Water use history survives firmware updates. From firmware 2.8 it is protected in three places:

Where	What it does
The controller	Stores the history twice (main record and backup copy), saves it again before every wireless update, and never overwrites a record it cannot read
Every phone that opens the app	Keeps its own copy, updated automatically. If the controller ever comes back without history, the Water use tab shows the phone's copy and a **Restore to controller** button
A history file	**Download history file** on the Water use tab. **Restore from file** brings it back, even onto a replaced or fully erased controller

If the Water use tab is empty after an update:

1. Open the app on a phone that was used before the update. Use the same address as usual: `greywater.local` and the IP address keep separate copies.
2. On the Water use tab, press **Restore to controller**. If no phone has a copy, use **Restore from file** with a downloaded history file.
3. Check the Service tab: *Water history* should read *Restored from the app*.

Avoid erasing the controller's memory: a normal upload keeps the history. A full flash erase removes it, together with the settings and the Wi-Fi network. That covers Burn Bootloader, recovery mode, uploading with another board or partition scheme selected, and Erase All Flash. Download a history file before any such work.

Safety and service features (firmware 2.9)

Feature	What happens	Where to see it
Interrupted cycle	If the power goes off, the program crashes or the watchdog restarts the controller during a cycle, the controller comes back in Ready with all outputs safe. It shows `INTERRUPTED` / *Check machine*, sends a high-priority alert and logs the event. The cycle is not resumed.	Screen, phone alert, Overview notice (until OK or the next Start), Service → Recent events
Watchdog	If the program freezes for 5 s, the controller restarts with every output off	Service → Last restart: *Watchdog (program froze)*
Wireless updates	Accepted only while the controller is Ready and no tank wash runs. During a cycle, the Arduino IDE's network upload gets no answer	Service → Wireless updates
Valves and pump	The Service tab shows what each output is actually set to	Service → Valves and pump
Event log	The last 50 events since the controller started: steps, machine filled, pump on and off with the reason, alerts, tank wash, Wi-Fi, updates, restores	Service → Recent events
Password protection	The app asks for new passwords while the defaults are in use. After 5 wrong passwords in a minute from one device, it refuses that device for a minute; other devices keep working (per device from 3.0)	App
Buzzer (optional)	An active buzzer module on D12, switched on in Settings: two short beeps when a cycle ends, three long beeps on alarms	Settings → Cycle

Feature	What happens	Where to see it
Typical step times	The controller learns how long steps 1–3 usually take; the app shows “Usually about 25 min”	Overview

A printable **Quick Card** for the household, with the installer’s commissioning checklist, is a separate document.

Firmware 3.0 and 3.1

Feature	What happens	Where to see it
Safety task (3.0)	Every 20 ms, independent of the app and Wi-Fi: reads the floats, stops the pump within about 0.3 s when the machine stops drawing, keeps the fill and tank-wash valves closed while the high float is wet	Service → Valves and pump; Recent events
Float-switch fault (3.0)	High float wet with the low float dry for 3 s: the tank is treated as full, the pump is not used, an alert is sent and the alarm beeps	Alert *check level switches*; app *Check float switches*; Service → Tank level switches
No screen (3.0)	If the screen does not answer at start-up, the controller runs without it and sends an alert	Service → Screen
Confirmed actions (3.0)	The app shows *Cycle started*, *Cycle stopped* or *Tank wash started* only after the controller confirms; *Send a test alert* reports whether the alert was delivered	App messages; Service → Last alert
Loop time and memory (3.0)	The slowest program pass and the free memory, to spot problems early	Service → Slowest loop pass, Memory

Feature	What happens	Where to see it
Language (3.1)	Settings → Language: Español (default) or English, for the app, the screen and the phone alerts. The event log stays in English for service	Settings

Spanish screen texts (3.1): the screen font has no accents, so the screen shows short plain words, for example `LISTO` / *Pulse START*, `1/4 LAVADO`, `2/4 DESAGUE`, `3/4 ENJUAGUE`, `4/4 AHORRO`, *Reusando agua*, *Agua de la llave*, *Guardando*, `TERMINADO`. The app shows the full Spanish sentences. The Wi-Fi setup page itself stays in English (it comes from the WiFiManager library).

Manual tank wash

Use it to rinse the storage tank between washes. The wash valve opens and closes by itself as soon as the high-level switch reports that the tank is full.

1. Check that the dashboard shows **Ready** and the tank is not full.
2. In the **Storage tank** panel, press **Wash tank**. The screen on the controller shows `TANK WASH` and the time left before the safety limit.
3. Wait. When the high-level switch reports water, the valve closes and the dashboard shows *Tank full: wash valve closed*.
4. To end it early, press **Stop washing** on the dashboard or **STOP** on the controller. Starting a cycle also ends it.

The wash ends because	The valve	Screen and dashboard	Phone alert
High-level switch reports full	Closes at once	*Tank full: valve closed*	Tank wash complete
High-level switch wet, low-level switch dry	Closes at once	*Level switches disagree: check sensors*	Check level switches (high priority)
Safety time limit reached (default 10 min)	Closes	*Time limit: check level switch*	Tank wash time limit (high priority)
Stop washing, STOP button, or a cycle starts	Closes	*Tank wash stopped*	None

The button is unavailable while a cycle runs or when the tank is already full.

Phone alerts

Install the free **ntfy** app, subscribe to a long, hard-to-guess topic name, enter the same topic in Settings and press **Send a test alert**.

Alert	Sent when	Priority
Cycle complete	Step 4 finishes; includes litres reused and saved	Low (silent on most phones)
Cycle stopped	STOP pressed on the controller or the dashboard	Minimum (silent)
Tank full	The tank fills during step 4 and extra water goes to the drain	Low
Check sensors	Step 1, 2 or 3 runs longer than the alert time (default 90 min) without the expected flow	High
Tank water is old	Stored water is older than the limit (default 48 h)	Normal
Tank wash complete	A manual tank wash filled the tank and the valve closed	Low
Check level switches	A manual wash stopped because the high-level switch is wet but the low-level switch is dry	High
Tank wash time limit	A manual wash reached its time limit before the tank reported full	High
Tank pump not pumping	The tank pump started 3 times in one cycle without any water flowing; tap water was used instead	High
Cycle interrupted	The controller restarted during a cycle (power cut, crash or watchdog)	High

Routine news is low priority so people don’t mute the topic; problems are high priority so they are noticed.

Alerts raised while Wi-Fi is down are sent if it returns within 60 seconds. Anyone who knows the topic name can read the alerts, so keep it private.

Wireless firmware updates

1. Stop any running cycle.
2. In the Arduino IDE, choose **Tools > Port** and the network port named `greywater`.
3. Upload and enter the update password.
4. The controller switches every valve and the pump off, shows `UPDATING`, then restarts.

Settings on the dashboard

Setting	Default	Allowed range
Wash every	10 s	1–600 s
Keep the valve open	2 s	0.25 s up to the interval minus 0.25 s
Manual wash time limit	10 min	1–30 min
Final drain length (step 4)	20 min	1–120 min
Flow detection time	4 s	1–30 s
Flow threshold	1 L/min	0.2–10 L/min; water counts as flowing above it and as stopped below it for 2 s
Minimum fill	3 L	0.5–50 L; water the machine must take before a drain can end Fill 1/4 or Rinse 3/4
Start from the dashboard	On	On / off
Buzzer	Off	On / off; needs an active buzzer module on D12
Warn when a step takes longer than	90 min	10–600 min

Setting	Default	Allowed range
Warn when tank water is older than	48 h	1–240 h
Alert server / topic	`https://ntfy.sh` / empty (alerts off)	*Create a private topic* fills in a random topic and shows the subscribe link
Time zone	UTC−5 (`<-05>5`)	Chosen from a list of common zones; *Other* accepts a POSIX code
Dashboard and update passwords	`greywater` (the app asks you to change it)	8–32 characters; the default cannot be chosen again; both take effect at once (from 3.0)

Wi-Fi troubleshooting

Problem	What to check	How to fix
`greywater.local` does not open	Phone on the same Wi-Fi; some Android phones do not support `.local` names	Use the IP address shown on the screen at start-up, or from the router's device list
Dashboard says it can't reach the controller	Controller powered; Wi-Fi signal (shown in the page footer)	Press Refresh in the top bar. If it stays Offline or Updating slowly, move the router or add a repeater; below about −80 dBm the link drops
No alerts arrive	Topic in Settings matches the app; footer says alerts are on	Press **Send a test alert**; check the phone's notification permission
Network changed	—	Hold STOP for 5 s while `READY`, join GreyWater-Setup with the key shown on the screen, and set the new network

Problem	What to check	How to fix
Wireless upload fails	Network port listed in the IDE; update password	Retry; if it keeps failing, upload once over USB
Water age shows *No water stored* with water in the tank	The tank level must be above the low switch for 30 s	Check the low-level switch

Security

- Change both default passwords on first use.
- Keep the dashboard on the home network; do not forward router ports to it.
- Turn **Start from the dashboard** off in Settings if a cycle could be started while nobody is at the machine.
- The Wi-Fi setup network uses a key unique to each board, shown on the screen while the setup page is open (from firmware 3.0).

Appendix A. Abbreviations

Term	Meaning
EV	Electrovalve (solenoid valve)
FSM	Finite state machine
GND	Ground
I2C	Two-wire serial bus used by the OLED (SDA data, SCL clock)
ISR	Interrupt service routine
OLED	Organic LED display
RCD / GFCI	Residual-current device / ground-fault circuit interrupter
SSR	Solid-state relay

Appendix B. Revision history

Version	Date	Description
3.1	2026-10-30	Spanish (Colombia) for the app, the screen and the phone alerts, chosen in Settings → Language (Spanish by default); alerts sent as JSON so accents arrive intact; setup key shown on the screen
3.0	2026-10-30	Safety task and output interlocks; float-switch rules, debouncing and fault alert; runs without a screen; per-board Wi-Fi setup key and no firmware upload on the setup page; update password applied at once; lockout per device; app reports only confirmed actions and alert delivery; compressed app page; code split into modules. Tested on the installed system
2.9	2026-10-29	Interrupted-cycle detection and alert; watchdog; wireless updates blocked during a cycle; real valve and pump states and an event log on the Service tab; password prompt and lockout; settings in the history file; time zone list; private alert topic; quieter routine alerts; typical step times; optional buzzer on D12; full-cycle simulation tests; Quick Card
2.8.1	2026-10-29	Refresh button in the app’s top bar; requests time out so a slow connection no longer freezes the screen; “Updating slowly” badge after 6 s without an update
2.8	2026-10-29	Water history is never lost: stored twice on the controller, unreadable records kept aside, saved before wireless updates; each phone keeps a copy and can restore it; history file download and restore; history source shown on the Service tab
2.7	2026-10-21	Controller screen rewritten for households, all messages fit without scrolling; app reorganised into Overview, Water use, Service and Settings tabs

Version	Date	Description
2.6	2026-10-21	“Machine is filled” guard (fixes the early move to Drain introduced in 2.4); Minimum fill setting; settings migrate without reset
2.5	2026-10-21	Tank pump protection: the pump stops within 0.3 s when the machine closes its inlet; dry-start alert
2.4	2026-10-21	Drain-sensor fault fixed: new flow meter; diode and pull-up sensor wiring; FlowSensor library removed; accessibility fixes
2.3	2026-10-21	Dashboard shows results: This month panel, last wash summary, tank readiness
2.2	2026-10-21	Manual tank wash that stops when the tank is full
2.1	2026-10-21	Redesigned dashboard with Start and Stop buttons; tank-wash valve moved to D0
2.0	2026-10-21	Wi-Fi: dashboard, phone alerts, statistics, wireless updates, settings
1.2	2026-10-19	Tank-wash valve moved from D12 to D1
1.1	2026-10-19	Quick start, pinout drawing, wiring diagrams, clearer tables
1.0	2026-10-19	Initial release